



数据库系统课程实验报告

实验名称:	数据库的安全性
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1. 实验目的

- 理解数据库系统用户(user)、权限(privilege)和角色(role)的概念和作用
- 熟练掌握用户的管理：创建、查看、删除和权限的授予与回收
- 熟练掌握通过数据字典查看用户权限、表和视图权限的方法
- 熟练掌握使用 Grant 命令给用户、角色授权的方法
- 熟练掌握使用 Revoke 命令回收已授权限的方法
- 熟练掌握角色定义、重命名和删除的方法
- 熟练掌握修改角色中权限的方法
- 理解视图的安全性作用

2. 实验内容和步骤

(1) 学 习 并 完 成

<https://blog.csdn.net/BlacKingZ/article/details/117912175> 的内容

步骤如下：

1. 创建用户 ks，登录密码为 KoShi641255

```
postgres=# CREATE USER ks PASSWORD 'KoShi641255';  
CREATE ROLE
```

2. 查看用户列表

```
postgres=# SELECT * FROM pg_user;  
username | usesysid | usecreatedb | usesuper | usecatupd | userepl | passwd | valbegin | valuntil | respool | parent | spacelimit | useconfig  
| nodegroup | tempaclimit | spillspaclimit | usemonitoradmin | useoperatoradmin | usepolicyadmin  
+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+-----+  
omm | 10 | t | t | t | t | ***** | | | default_pool | 0 | |  
| | | | | | | | | | | | |  
mx | 16579 | t | f | f | f | ***** | | | default_pool | 0 | |  
| | | | | | | | | | | | |  
ks | 16825 | f | f | f | f | ***** | | | default_pool | 0 | |  
| | | | | | | | | | | | |  
(3 rows)
```

3. 为用户 ks 追加有创建角色的 CREATE ROLE 权限

```
postgres=# ALTER USER ks CREATEROLE;  
ALTER ROLE
```

4.删除用户

```
postgres=# DROP USER ks CASCADE;  
DROP ROLE
```

后续实验需要收费观看，略。

(2) 创建视图 salesman，该视图只保存 employees 表中所有 job_title 为'Sales Representative'的雇员

步骤如下：

```
sales=> CREATE VIEW salesman AS SELECT * FROM employees WHERE job_title='Sales Representative';  
CREATE VIEW
```

(3) 创建基于 salesman 的视图 salesman_contacts(first_name,last_name,email,phone)，该视图存储的 salesman 的联系方式

步骤如下：

```
sales=> CREATE VIEW salesman_contacts AS SELECT first_name,last_name,email,phone FROM salesman;  
CREATE VIEW
```

(4) 查询视图 salesman 和 salesman_contacts

步骤如下：

1.查询视图 salesman

```
sales=> SELECT * FROM salesman;
```

employee_id	first_name	last_name	email	phone	hire_date	manager_id	job_title
51	Poppy	Jordan	poppyjordan@examplecom	1.14413E+13	2030-09-16 00:00:00	46	Sales Representative
52	Sophie	Owens	sophieowens@examplecom	1.14413E+13	2024-03-16 00:00:00	46	Sales Representative
53	Sophia	Reynolds	sophiareynolds@examplecom	1.14413E+13	2020-09-16 00:00:00	46	Sales Representative
54	Lily	Fisher	lilyfisher@examplecom	1.14413E+13	2030-03-16 00:00:00	46	Sales Representative
55	Grace	Ellis	graceellis@examplecom	1.14413E+13	2009-12-16 00:00:00	46	Sales Representative
56	Evie	Harrison	evieharrison@examplecom	1.14413E+13	2023-11-16 00:00:00	46	Sales Representative
57	Scarlett	Gibson	scarlettgibson@examplecom	1.14413E+13	2030-09-16 00:00:00	47	Sales Representative
58	Ruby	Mcdonald	rubymcdonald@examplecom	1.14413E+13	2004-03-16 00:00:00	47	Sales Representative
59	Chloe	Cruz	chloecruz@examplecom	1.14413E+13	2001-09-16 00:00:00	47	Sales Representative
60	Isabelle	Marshall	isabellemarshall@examplecom	1.14413E+13	2010-03-16 00:00:00	47	Sales Representative
61	Daisy	Ortiz	daisyortiz@examplecom	1.14413E+13	2015-12-16 00:00:00	47	Sales Representative
62	Freya	Gomez	freyagomez@examplecom	1.14413E+13	2003-11-16 00:00:00	47	Sales Representative
63	Phoebe	Murray	phoebemurray@examplecom	1.14413E+13	2011-11-16 00:00:00	48	Sales Representative
64	Florence	Freeman	florencefreeman@examplecom	1.14413E+13	2019-03-16 00:00:00	48	Sales Representative
65	Alice	Wells	alicewells@examplecom	1.14413E+13	2024-09-16 00:00:00	48	Sales Representative
66	Charlotte	Webb	charlottewebb@examplecom	1.14413E+13	2023-02-16 00:00:00	48	Sales Representative
67	Sienna	Simpson	siennasimpson@examplecom	1.14413E+13	2024-03-16 00:00:00	48	Sales Representative
68	Matilda	Stevens	matildastevens@examplecom	1.14413E+13	2021-02-16 00:00:00	48	Sales Representative
69	Evelyn	Tucker	evelyntucker@examplecom	1.14413E+13	2011-03-16 00:00:00	49	Sales Representative
70	Eva	Porter	evaporter@examplecom	1.14413E+13	2023-03-16 00:00:00	49	Sales Representative
71	Millie	Hunter	milliehunter@examplecom	1.14413E+13	2024-09-16 00:00:00	49	Sales Representative
72	Sofia	Hicks	sofiahicks@examplecom	1.14413E+13	2023-02-16 00:00:00	49	Sales Representative
73	Lucy	Crawford	lucycrawford@examplecom	1.14413E+13	2024-03-16 00:00:00	49	Sales Representative
74	Elsie	Henry	elsiehenry@examplecom	1.14413E+13	2021-02-16 00:00:00	49	Sales Representative
75	Imogen	Boyd	imogenboyd@examplecom	1.14416E+13	2011-05-16 00:00:00	50	Sales Representative
76	Layla	Mason	laylamason@examplecom	1.14416E+13	2019-03-16 00:00:00	50	Sales Representative
77	Rosie	Morales	rosiemorales@examplecom	1.14416E+13	2024-03-16 00:00:00	50	Sales Representative
78	Maya	Kennedy	mayakennedy@examplecom	1.14416E+13	2023-02-16 00:00:00	50	Sales Representative
79	Esme	Warren	esmewarren@examplecom	1.14416E+13	2024-05-16 00:00:00	50	Sales Representative
80	Elizabeth	Dixon	elizabethdixon@examplecom	1.14416E+13	2004-09-16 00:00:00	50	Sales Representative

(30 rows)

2. 查询视图 salesman_contacts

```

sales=> SELECT * FROM salesman_contacts;
first_name | last_name | email | phone
-----+-----+-----+-----
Poppy | Jordan | poppyjordan@examplecom | 1.14413E+13
Sophie | Owens | sophieowens@examplecom | 1.14413E+13
Sophia | Reynolds | sophiareynolds@examplecom | 1.14413E+13
Lily | Fisher | lilyfisher@examplecom | 1.14413E+13
Grace | Ellis | graceellis@examplecom | 1.14413E+13
Evie | Harrison | evieharrison@examplecom | 1.14413E+13
Scarlett | Gibson | scarlettgibson@examplecom | 1.14413E+13
Ruby | Mcdonald | rubymcdonald@examplecom | 1.14413E+13
Chloe | Cruz | chloecruz@examplecom | 1.14413E+13
Isabelle | Marshall | isabellemarshall@examplecom | 1.14413E+13
Daisy | Ortiz | daisyortiz@examplecom | 1.14413E+13
Freya | Gomez | freyagomez@examplecom | 1.14413E+13
Phoebe | Murray | phoebemurray@examplecom | 1.14413E+13
Florence | Freeman | florencefreeman@examplecom | 1.14413E+13
Alice | Wells | alicewells@examplecom | 1.14413E+13
Charlotte | Webb | charlottewebb@examplecom | 1.14413E+13
Sienna | Simpson | siennasimpson@examplecom | 1.14413E+13
Matilda | Stevens | matildastevens@examplecom | 1.14413E+13
Evelyn | Tucker | evelyntucker@examplecom | 1.14413E+13
Eva | Porter | evaporter@examplecom | 1.14413E+13
Millie | Hunter | milliehunter@examplecom | 1.14413E+13
Sofia | Hicks | sofiahicks@examplecom | 1.14413E+13
Lucy | Crawford | lucycrawford@examplecom | 1.14413E+13
Elsie | Henry | elsiehenry@examplecom | 1.14413E+13
Imogen | Boyd | imogenboyd@examplecom | 1.14416E+13
Layla | Mason | laylamason@examplecom | 1.14416E+13
Rosie | Morales | rosiemorales@examplecom | 1.14416E+13
Maya | Kennedy | mayakennedy@examplecom | 1.14416E+13
Esme | Warren | esmewarren@examplecom | 1.14416E+13
Elizabeth | Dixon | elizabethdixon@examplecom | 1.14416E+13
(30 rows)

```

(5) 在当前窗口输入命令：\c - omm 切换到 omm 用户

步骤如下：

```

sales=> \c - omm
Non-SSL connection (SSL connection is recommended when requiring high-security)
You are now connected to database "sales" as user "omm".

```

(6) 创建新用户 user1

步骤如下:

```
sales=# CREATE USER user1 PASSWORD 'User114514';  
CREATE ROLE
```

(7) 在当前窗口输入命令: \c - user1 切换到 user1 用户

步骤如下:

```
sales=# \c - user1  
Password for user user1:  
Non-SSL connection (SSL connection is recommended when requiring high-security)  
You are now connected to database "sales" as user "user1".
```

(8) 发布查询命令: select * from salesman_contacts;观察结果

步骤如下:

```
sales=> SELECT * FROM salesman_contacts;  
ERROR: relation "salesman_contacts" does not exist on dn_6001  
LINE 1: SELECT * FROM salesman_contacts;
```

(9) 发布命令:\c - whj 切换到 whj 用户——此处的 whj 应替换为你们自己创建的用户 (下同)

步骤如下:

```
sales=> \c - mx  
Password for user mx:  
Non-SSL connection (SSL connection is recommended when requiring high-security)  
You are now connected to database "sales" as user "mx".
```

同时需要切换到相应模式下

```
sales=> \c Sales;  
Password for user mx:  
Non-SSL connection (SSL connection is recommended when requiring high-security)  
You are now connected to database "sales" as user "mx".
```

```
sales=> SET SEARCH_PATH TO Sales,public;  
SET
```

(10) 在当前 whj 用户下输入命令: grant select on salesman_contacts to user1; 实现授权操作。

步骤如下:

```
sales=> GRANT SELECT ON salesman_contacts TO user1;  
GRANT
```

(11) 依次重复步骤 (7) 和 (8) , 比较两次查询的结果

/*说明: 步骤 (2) - (11) 的主要目的是用于验证视图的作用: 被授权用户只能查询在权限范围内的数据, 范围外的数据不可访问*/

步骤如下:

1.要想 user1 也能访问 salesman_contacts, 必须先给其模式 Sales 的访问权限。

```
sales=> GRANT USAGE ON SCHEMA Sales TO user1;  
GRANT
```

2.切换到用户 user1 进行查询。

```
sales=> \c - user1;  
Password for user user1:  
Non-SSL connection (SSL connection is recommended when requiring high-security)  
You are now connected to database "sales" as user "user1".
```

```
sales=> \c Sales;  
Password for user user1:  
Non-SSL connection (SSL connection is recommended when requiring high-security)  
You are now connected to database "sales" as user "user1".
```

```
sales=> SET SEARCH_PATH TO Sales,public;  
SET
```

```
sales=> SELECT * FROM salesman_contacts;
```

first_name	last_name	email	phone
Poppy	Jordan	poppyjordan@examplecom	1.14413E+13
Sophie	Owens	sophieowens@examplecom	1.14413E+13
Sophia	Reynolds	sophiareynolds@examplecom	1.14413E+13
Lily	Fisher	lilyfisher@examplecom	1.14413E+13
Grace	Ellis	graceellis@examplecom	1.14413E+13
Evie	Harrison	evieharrison@examplecom	1.14413E+13
Scarlett	Gibson	scarlettgibson@examplecom	1.14413E+13
Ruby	Mcdonald	rubymcdonald@examplecom	1.14413E+13
Chloe	Cruz	chloecruz@examplecom	1.14413E+13
Isabelle	Marshall	isabellemarshall@examplecom	1.14413E+13
Daisy	Ortiz	daisyortiz@examplecom	1.14413E+13
Freya	Gomez	freitagomez@examplecom	1.14413E+13
Phoebe	Murray	phoebemurray@examplecom	1.14413E+13
Florence	Freeman	florencefreeman@examplecom	1.14413E+13
Alice	Wells	alicewells@examplecom	1.14413E+13
Charlotte	Webb	charlottewebb@examplecom	1.14413E+13
Sienna	Simpson	siennasimpson@examplecom	1.14413E+13
Matilda	Stevens	matildastevens@examplecom	1.14413E+13
Evelyn	Tucker	evelyntucker@examplecom	1.14413E+13
Eva	Porter	evaporter@examplecom	1.14413E+13
Millie	Hunter	milliehunter@examplecom	1.14413E+13
Sofia	Hicks	sofiahicks@examplecom	1.14413E+13
Lucy	Crawford	lucycrawford@examplecom	1.14413E+13
Elsie	Henry	elsiehenry@examplecom	1.14413E+13
Imogen	Boyd	imogenboyd@examplecom	1.14416E+13
Layla	Mason	laylamason@examplecom	1.14416E+13
Rosie	Morales	rosiemorales@examplecom	1.14416E+13
Maya	Kennedy	mayakennedy@examplecom	1.14416E+13
Esme	Warren	esmewarren@examplecom	1.14416E+13
Elizabeth	Dixon	elizabethdixon@examplecom	1.14416E+13

(30 rows)

(12) 查看与角色、权限相关的系统表和系统视图:pg_roles,pg_authid

步骤如下:

1.切换到超级用户。

```
sales=> \c - omm
Non-SSL connection (SSL connection is recommended when requiring high-security)
You are now connected to database "sales" as user "omm".
```

2.查看 pg_roles。


```
sc=> GRANT SELECT ON TABLE SC TO PUBLIC;  
GRANT
```

4.把查询 Student 表和修改学生学号的权限授给用户 U4。

```
sc=> GRANT UPDATE(Sno),SELECT ON TABLE Student TO U4;  
GRANT
```

5.把对表 SC 的 INSERT 权限授予 U5 用户，并允许将此权限再授予其他用户。

```
sc=> GRANT INSERT ON TABLE SC TO U5 WITH GRANT OPTION;  
GRANT
```

6.U5 将对表 SC 的 INSERT 权限授予 U6 用户，并允许将此权限再授予其他用户。

```
Previous connection kept  
sc=> \c - U5  
Password for user u5:  
Non-SSL connection (SSL connection is recommended when requiring high-security)  
You are now connected to database "sc" as user "u5".
```

```
Previous connection kept  
sc=> \c - U5  
Password for user u5:  
Non-SSL connection (SSL connection is recommended when requiring high-security)  
You are now connected to database "sc" as user "u5".
```

```
sc=> SET SEARCH_PATH TO SC,public;  
SET  
sc=> GRANT INSERT ON TABLE SC TO U6 WITH GRANT OPTION;  
GRANT
```

7.U6 将对表 SC 的 INSERT 权限授予 U7 用户。

```
sc=> GRANT USAGE ON SCHEMA SC TO U6;  
GRANT
```

```
sc=> \c - U6  
Password for user u6:  
Non-SSL connection (SSL connection is recommended when requiring high-security)  
You are now connected to database "sc" as user "u6".  
sc=> SET SEARCH_PATH TO SC,public;  
SET  
sc=> GRANT INSERT ON TABLE SC TO U7;  
GRANT
```

8.把用户 U4 修改学生学号的权限收回。

```

sc=> \c - mx
Password for user mx:
Non-SSL connection (SSL connection is recommended when requiring high-security)
You are now connected to database "sc" as user "mx".
sc=> SET SEARCH_PATH TO SC,public;
SET
sc=> REVOKE UPDATE(Sno) ON TABLE Student FROM U4;
REVOKE

```

9.收回所有用户对表 SC 的查询权限。

```

sc=> REVOKE SELECT ON TABLE SC FROM PUBLIC;
REVOKE

```

10.把用户 U5 对 SC 表的 INSERT 权限收回。

```

sc=> REVOKE INSERT ON TABLE SC FROM U5 CASCADE;
REVOKE

```

11. 建立计算机系学生的视图，把对该视图的 SELECT 权限授予王平，把该视图上的所有操作权限授予张明。

由于中文名出现 invalid name，采用拼音命名。

```

sc=> CREATE VIEW CS_Student AS SELECT * FROM Student WHERE Sdept='CS';
CREATE VIEW

```

```

sc=> GRANT ALL PRIVILEGES ON CS_Student TO zhangming;
GRANT

```

```

sc=> GRANT SELECT ON CS_Student TO wangping;
GRANT

```

具体的验证过程略

3.实验总结

3.1 完成的工作

根据实验内容，完成对用户以及权限的管理操作。

3.2 对实验的认识

本实验包含对数据库安全性的操作，并含有对表的基本操作的复习，是很好的练习机会。通过操作 openGauss 数据库，能够体会到其

与 SQL 语句在安全操作上的些许区别。

思考题：

具有什么权限才能创建新用户？

答：超级用户 omm 或者具有 CREATEROLE 权限的用户。

角色的作用是什么？

答：身份认证和访问控制、权限分组和管理、继承和层次结构、简化权限管理。

如何实现角色所含权限的修改，请设计样例验证之。

答：具体案例可参考实验（13），权限修改主要围绕两个关键字 GRANT 和 REVOKE 进行实现。

3.3 遇到的困难及解决方法

由于基本表是在模式下的，在切换用户时，需要重新设置模式的路径已经授予模式访问的权限。审计功能似乎很难具体到某个表的操作，目前暂未实现，其 `gs_guc` 命令和 `CREATE AUDIT` 命令与所学 SQL 语句大不相同。